

BanglaSpeechFM: Towards a Foundation Model for Bangla Speech

Anonymous ACL submission

Abstract

The rapid advancement of self-supervised learning (SSL) has revolutionized speech representation, yet low-resource languages like Bengali remain largely dependent on fine-tuning massive, computationally expensive multilingual foundation models. This reliance not only imposes prohibitive resource constraints but also frequently overlooks the unique phonetic nuances of the Bengali language. To address this critical gap, this thesis introduces a Continuous Latent Autoencoder (CLAE)—a highly efficient, 23.8M-parameter speech foundation model developed entirely from scratch for 16 kHz Bengali speech. Trained exclusively on publicly available datasets using a novel self-supervised objective that optimally combines mel-spectrogram reconstruction, Joint-Embedding Predictive Architecture (JEPA), and Variance-Invariance-Covariance Regularization (VISReg), our approach generates robust, frame-level representations at a 12.5 Hz latent sequence rate. Despite its remarkably compact footprint—featuring a 15.29M-parameter frozen feature extractor—the proposed model achieves highly competitive, and in specific evaluation metrics, superior performance compared to state-of-the-art heavyweight speech models. This research breaks new ground by unequivocally demonstrating that training a dedicated, lightweight foundation model from scratch is not only a viable pathway for low-resource speech AI but also yields highly effective representations for downstream analysis. By providing a scalable, resource-efficient architecture, this work establishes a new paradigm for accessible and specialized speech representation learning in Bengali, paving the way for future innovations in low-resource linguistic technologies.

1 Introduction

These instructions are for authors submitting papers to *ACL conferences using L^AT_EX. They are

not self-contained. All authors must follow the general instructions for *ACL proceedings,¹ and this document contains additional instructions for the L^AT_EX style files.

The templates include the L^AT_EX source of this document (acl_latex.tex), the L^AT_EX style file used to format it (acl.sty), an ACL bibliography style (acl_natbib.bst), an example bibliography (custom.bib), and the bibliography for the ACL Anthology (anthology.bib).

2 Engines

To produce a PDF file, pdfL^AT_EX is strongly recommended (over original L^AT_EX plus dvips+ps2pdf or dvipdf). The style file acl.sty can also be used with luaL^AT_EX and XeL^AT_EX, which are especially suitable for text in non-Latin scripts. The file acl_lualatex.tex in this repository provides an example of how to use acl.sty with either luaL^AT_EX or XeL^AT_EX.

3 Preamble

The first line of the file must be

```
\documentclass[11pt]{article}
```

To load the style file in the review version:

```
\usepackage[review]{acl}
```

For the final version, omit the review option:

```
\usepackage{acl}
```

To use Times Roman, put the following in the preamble:

```
\usepackage{times}
```

¹<http://acl-org.github.io/ACLPUb/formatting.html>

Command	Output	Command	Output
<code>\a</code>	ä	<code>\c c</code>	ç
<code>\^e</code>	ê	<code>\u g</code>	ğ
<code>\`i</code>	ì	<code>\l</code>	ł
<code>\.I</code>	İ	<code>\~n</code>	ñ
<code>\o</code>	ø	<code>\H o</code>	ö
<code>\'u</code>	ú	<code>\v r</code>	ř
<code>\aa</code>	å	<code>\ss</code>	ß

Table 1: Example commands for accented characters, to be used in, *e.g.*, Bib_{TEX} entries.

(Alternatives like `txfonts` or `newtx` are also acceptable.)

Please see the $\LaTeX$ source of this document for comments on other packages that may be useful.

Set the title and author using `\title` and `\author`. Within the author list, format multiple authors using `\and` and `\And` and `\AND`; please see the $\LaTeX$ source for examples.

By default, the box containing the title and author names is set to the minimum of 5 cm. If you need more space, include the following in the preamble:

```
\setlength\titlebox{<dim>}
```

where `<dim>` is replaced with a length. Do not set this length smaller than 5 cm.

4 Document Body

4.1 Footnotes

Footnotes are inserted with the `\footnote` command.²

4.2 Tables and figures

See Table 1 for an example of a table and its caption. **Do not override the default caption sizes.**

As much as possible, fonts in figures should conform to the document fonts. See Figure 1 for an example of a figure and its caption.

Using the `graphicx` package `graphics` files can be included within figure environment at an appropriate point within the text. The `graphicx` package supports various optional arguments to control the appearance of the figure. You must include it explicitly in the $\LaTeX$ preamble (after the `\documentclass` declaration and before `\begin{document}`) using `\usepackage{graphicx}`.

²This is a footnote.



Figure 1: A figure with a caption that runs for more than one line. Example image is usually available through the `mwe` package without even mentioning it in the preamble.

4.3 Hyperlinks

Users of older versions of $\LaTeX$ may encounter the following error during compilation:

```
\pdfendlink ended up in different nest-
ing level than \pdfstartlink.
```

This happens when `pdf $\LaTeX$` is used and a citation splits across a page boundary. The best way to fix this is to upgrade $\LaTeX$ to 2018-12-01 or later.

4.4 Citations

Table 2 shows the syntax supported by the style files. We encourage you to use the `natbib` styles. You can use the command `\citet` (cite in text) to get “author (year)” citations, like this citation to a paper by [Gusfield \(1997\)](#). You can use the command `\citep` (cite in parentheses) to get “(author, year)” citations ([Gusfield, 1997](#)). You can use the command `\citealp` (alternative cite without parentheses) to get “author, year” citations, which is useful for using citations within parentheses (e.g. [Gusfield, 1997](#)).

A possessive citation can be made with the command `\citeposs`. This is not a standard `natbib` command, so it is generally not compatible with other style files.

4.5 References

The $\LaTeX$ and Bib_{TEX} style files provided roughly follow the American Psychological Association format. If your own bib file is named `custom.bib`, then placing the following before any appendices in your $\LaTeX$ file will generate the references section for you:

```
\bibliography{custom}
```

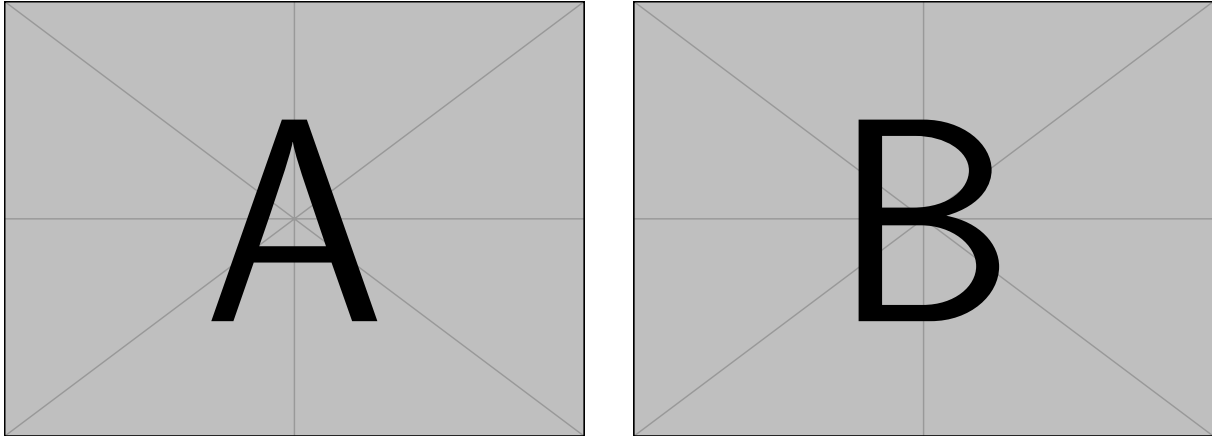


Figure 2: A minimal working example to demonstrate how to place two images side-by-side.

Output	natbib command	ACL only command
(Gusfield, 1997)	<code>\citep</code>	
Gusfield, 1997	<code>\citealp</code>	
Gusfield (1997)	<code>\citet</code>	
(1997)	<code>\citeyearpar</code>	
Gusfield’s (1997)		<code>\citeposs</code>

Table 2: Citation commands supported by the style file. The style is based on the natbib package and supports all natbib citation commands. It also supports commands defined in previous ACL style files for compatibility.

You can obtain the complete ACL Anthology as a BibTeX file from <https://aclweb.org/anthology/anthology.bib.gz>. To include both the Anthology and your own .bib file, use the following instead of the above.

`\bibliography{anthology,custom}`

Please see Section 5 for information on preparing BibTeX files.

4.6 Equations

An example equation is shown below:

$$A = \pi r^2 \tag{1}$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command.

This an example cross-reference to Equation 1.

4.7 Appendices

Use `\appendix` before any appendix section to switch the section numbering over to letters. See Appendix A for an example.

5 BibTeX Files

Unicode cannot be used in BibTeX entries, and some ways of typing special characters can disrupt BibTeX’s alphabetization. The recommended way of typing special characters is shown in Table 1.

Please ensure that BibTeX records contain DOIs or URLs when possible, and for all the ACL materials that you reference. Use the `doi` field for DOIs and the `url` field for URLs. If a BibTeX entry has a URL or DOI field, the paper title in the references section will appear as a hyperlink to the paper, using the `hyperref` LaTeX package.

Limitations

This document does not cover the content requirements for ACL or any other specific venue. Check the author instructions for information on maximum page lengths, the required “Limitations” section, and so on.

Acknowledgments

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Roskovskaya, ACL 2018 by Shay Cohen, Kevin Gimpel, and Wei Lu, NAACL 2018 by Margaret Mitchell and Stephanie Lukin, Bib_T_E_X suggestions for (NA)ACL 2017/2018 from Jason Eisner, ACL 2017 by Dan Gildea and Min-Yen Kan, NAACL 2017 by Margaret Mitchell, ACL 2012 by Maggie Li and Michael White, ACL 2010 by Jing-Shin Chang and Philipp Koehn, ACL 2008 by Johanna D. Moore, Simone Teufel, James Allan, and Sadaoki Furui, ACL 2005 by Hwee Tou Ng and Kemal Oflazer, ACL 2002 by Eugene Charniak and Dekang Lin, and earlier ACL and EACL formats written by several people, including John Chen, Henry S. Thompson and Donald Walker. Additional elements were taken from the formatting instructions of the *International Joint Conference on Artificial Intelligence* and the *Conference on Computer Vision and Pattern Recognition*.

References

- Rie Kubota Ando and Tong Zhang. 2005. A framework for learning predictive structures from multiple tasks and unlabeled data. *Journal of Machine Learning Research*, 6:1817–1853.
- Galen Andrew and Jianfeng Gao. 2007. Scalable training of L1-regularized log-linear models. In *Proceedings of the 24th International Conference on Machine Learning*, pages 33–40.
- Dan Gusfield. 1997. *Algorithms on Strings, Trees and Sequences*. Cambridge University Press, Cambridge, UK.
- Mohammad Sadegh Rasooli and Joel R. Tetreault. 2015. [Yara parser: A fast and accurate dependency parser](#). *Computing Research Repository*, arXiv:1503.06733. Version 2.

A Example Appendix

This is an appendix.